

An AutoML Platform for Environmental Agencies: An Approach to Transferable Modeling Using Air Pollutant Forecasting as a Case Study

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Abstract. Given global ecological challenges, environmental agencies require reliable, data-driven decision-making foundations. However, the increasing volume and heterogeneity of environmental data complicates their analysis. Machine learning (ML) offers methodological support for this purpose, yet its application remains limited in environmental administrations due to a lack of ML expertise. Within the Simplex4Learning (S4L) project, an AutoML platform was developed that combines methods of Automated Machine Learning (AutoML) and Explainable AI (XAI) with efficient approaches for (geo-)data integration and management. This paper outlines the platform in its current state of development and demonstrates its functionality using a concrete use case from environmental administration: the prediction of air pollutants for scenario analysis. We describe the application of the integrated technologies – from data provision and preparation through feature engineering to modeling and the explainability of learning outcomes. The aim of this paper is to illustrate the benefits of the S4L platform for professionals in environmental agencies, to demonstrate its transferable modeling capabilities across, for example, different federal states, and to discuss its practical suitability for administrative use.

Keywords: Machine Learning Platform, AutoML, Domain Expert Support, Environmental Data Analysis, Environmental Agencies, Transferability

1 Introduction

Faced with global environmental challenges such as climate change, biodiversity loss, and resource scarcity, the demand for reliable, data-driven decision-making in environmental agencies is growing. Despite extensive and continuously expanding environmental datasets from remote sensing, meteorology, and emissions monitoring, analysis remains complex due to data heterogeneity and distribution. While machine learning (ML) has established itself as an effective analytical method, its adoption in

public administration remains insufficient – partly owing to limited methodological expertise among domain users. Automated Machine Learning (AutoML) has the potential to lower these entry barriers by automating key steps in model development. Successful ML application in environmental agencies further requires user-friendly systems, professional support, and interdisciplinary collaboration.

In the research project Simplex4Learning (S4L, www.simplex4learning.de), an ML platform was developed to enable environmental agency professionals without deep ML expertise to conduct domain-specific ML analyses. Its architecture is oriented toward the standardized CRISP-DM process, adapted to the workflows of the participating agencies.

The paper outlines the platform in its current state, focusing particularly on the challenge of transferable modeling. This involves enabling analyses developed once to be applied to changing contexts without extensive manual adaptation, such as different pollutants within the same domain or identical environmental questions across different federal states. To validate this generic approach, we demonstrate the platform's functionality through air pollutant prediction for scenario analysis in two federal states, covering the full workflow from data provision and preparation through feature engineering and modeling to the explainability of results, and reflecting on the practical suitability of these automated workflows for administrative use.

The paper is structured as follows: Section 2 covers technical foundations and the state of research; Section 3 presents the S4L platform architecture; Section 4 describes the ML experiment methodology; Section 5 demonstrates the platform using an environmental administration use case; Section 6 discusses practical suitability; and Section 7 summarizes findings and provides an outlook on future work.

2 Background and State of research

The S4L platform represents an AutoML solution designed specifically for environmental agencies. Section 2.1 establishes the theoretical foundations by outlining the necessary base technologies, followed by Section 2.2, which discusses the state of research on ML platforms in this specific application domain. Section 2.3 then addresses the state of research on air quality data analysis using machine learning, providing the basis for the use case of pollutant prediction for scenario analysis.

2.1 Background

The S4L approach is based on established technical concepts to efficiently manage the ML lifecycle. To limit project complexity, the implementation focuses on three key areas: AutoML serves as the foundation for automating the development cycle from data preparation to model optimization [1–3]. MLOps scales the process from development to production monitoring through standardized pipelines [4]. For transparency in model decisions, Explainable AI (XAI) methods are employed to enable error diagnosis and validation [5]. Within XAI, LIME approximates black-box mod-

els using local, interpretable models [6], while SHAP provides a model-agnostic explanation for complex architectures [7].

Designed as a practice-oriented prototype for the German environmental administration, the S4L platform is based on two established base tools that are used in environmental administration. Disy Cadenza serves as a platform for Business & Location Intelligence, which acts as a link between existing spatial data infrastructures and the project through the integrated processing of attribute and spatial data [8]. In addition, the Simplex Twin Information System (SimplexTwIS) forms a data warehouse optimized for environmental data, which provides a harmonized data structure. It enables the realization of central data processing processes for the integration, harmonization, and enrichment of information [9, 10].

2.2 State of Research: ML Platforms for Environmental Agencies

For the development of classical ML methods – such as regression, classification, clustering, and dimensionality reduction – the Python framework Scikit-learn has established itself as the standard [11]. Although the framework is explicitly designed for non-specialists, its effective application in practice requires solid programming skills as well as a thorough understanding of ML methodology and statistical-mathematical concepts [12]. Furthermore, specific environmental datasets frequently require tailored preprocessing steps that go beyond standard functionalities.

To reduce barriers for users unfamiliar with machine learning, AutoML platforms, low/no-code system systems automate time-intensive steps in the machine learning lifecycle. For instance, H2O AutoML focuses on a fully automated process chain from feature engineering through to the training of ensemble models [13], while the KNIME Analytics Platform enables the intuitive creation of ML pipelines through visual programming. Nevertheless, even with these systems, a fundamental level of ML expertise remains indispensable [14].

Although these approaches simplify the modeling process, they have limitations in the context of public environmental administration. Effective use in government agencies requires continuous workflow support throughout the entire ML lifecycle. This includes the preselection of suitable datasets, data enrichment, well-founded decision support for model architecture and selection, as well as methodological support for model validation through Explainable AI (XAI).

While individual pilot projects exist for use in government agencies, such as the work of the AI Lab of the German Federal Environment Agency (UBA) [15], these are tailored to specific use cases and therefore cannot be easily transferred to other scenarios. Due to the lack of generalized machine learning workflows, the application of machine learning in environmental administration often remains limited to isolated pilot projects [16].

In summary, common approaches lack essential functions for effective use in environmental agencies, which is the foundation for the S4L developments. This concerns in particular the compatibility with standards-compliant governmental data and software infrastructures, native support for the joint processing of factual, sensor, and geospatial data, as well as an intuitive graphical user interface for domain experts

without specialized ML knowledge, while simultaneously providing interfaces for ML experts. Furthermore, productive operation requires continuous support of the entire ML lifecycle, automation potential for increasing efficiency, as well as mechanisms for explainability and transparency in order to ensure acceptance within official processes.

2.3 State of Research: Machine Learning for the Analysis of Air Quality Data

Modeling and predicting air pollutants involves navigating the interplay between stochastic emission processes, non-linear atmospheric reactions, and chaotic interactions of meteorology and topography [17, 18].

Purely data-driven methods often face challenges in practice regarding transferability, underlying data quality, and physical plausibility. A central challenge in air quality modeling is its strong locality. Models learn statistical patterns from historical time series of a specific monitoring station, but are generally unable to transfer these in a straightforward manner to other locations [19–21]

To address gaps in input data, imputation methods are often used in machine learning. However, in air quality applications, these methods can even degrade prediction quality compared to omitting the affected time periods [22]. Consequently, hybrid data fusion approaches have been developed that combine traffic data, land use, and meteorological models in order to compute pollutant maps at street-level resolution [23]. The extrapolation capabilities of many ML approaches are also problematic when applied to air quality data. If future meteorological conditions or emission scenarios lie outside the training range, prediction accuracy can collapse [24, 25]. In recent years, hybrid approaches have therefore been gaining in importance, as they are able to integrate physical or statistical boundary conditions directly into the model architecture, which significantly improves physical plausibility and extrapolation capability [26].

In summary, the modelling of air pollutant concentrations represents a complex challenge that requires domain-specific expert knowledge in addition to data-driven approaches. Nevertheless, ML models can be effectively employed for specific local questions, provided that location-specific characteristics are adequately integrated into the modelling process.

The focus of this article is not on generating new insights into ML-based air quality analysis. Rather, the aim is to demonstrate how the S4L platform supports the use of ML in government agencies for complex tasks. An important consideration is the synergistic collaboration between domain and machine learning experts. This collaboration is essential to successfully transfer scientifically grounded model development into administrative practice and ensure scalable use. Furthermore, the platform promotes the reuse of already developed machine learning models, blueprints, and pipelines, thereby enhancing both work efficiency and result quality.

3 The Simplex4Learning Platform: Architecture and Methodology

The S4L platform was designed as a loosely coupled architecture of specialized software tools in order to make the use of machine learning (ML) in environmental administration both efficient and user-friendly. Through the integration of core components such as AutoML, Explainable AI (XAI), and MLOps, the platform enables continuous process support – from data preparation through model development to operational deployment. This approach aims to facilitate both the reuse of existing ML artefacts and the development of new models for novel research questions.

3.1 Conceptual Framework and Role Model

Within the scope of the S4L project, a role concept for the use of ML in large government agencies was developed, which defines responsibilities for the establishment, operation, and use of the infrastructure [27, 28]. A primary distinction is drawn between domain experts and data scientists. Domain experts possess in-depth specialist knowledge and focus on the substantive problem, the application and inference of models, as well as the interpretation of results, while data scientists contribute technical expertise in ML, statistics, and programming in order to guide data analysis and model development.

Based on this distribution of roles, two essential use cases can be distinguished in the application of the platform. The first case involves the “reuse” of existing solutions. Given that many machine learning problems in environmental agencies recur, existing models can be retrieved from the S4L library and efficiently adapted to new datasets or contexts. As the operational phase progresses, it can be expected that the number of supported use cases will steadily increase. This scenario is primarily aimed at domain experts to automate workflows with minimal technical effort. The second scenario involves the methodological development of new solutions, termed “New Problem”. When novel questions arise, boundary conditions change, or existing models prove insufficient, a complete ML process must be executed, a task typically performed by data scientists.

3.2 System Architecture and Methodological Approach

The technical architecture of the S4L platform integrates various components in order to implement the aforementioned scenarios, with the role concept defining the functions provided in each case. MLflow is employed for managing the ML lifecycle, serving as a central registry for models including history, versioning, and status management, as well as providing model comparison capabilities. The AutoML training environment integrates frameworks such as PyCaret for symbolic learning methods and AutoGluon for neural networks, thereby realising low-code and no-code approaches. The user interface is designed with specific roles in mind. Domain experts are seamlessly embedded into existing analysis and visualization workflows via the

platform “Disy Cadenza.” Data scientists, on the other hand, interact with the processes directly through Jupyter Notebooks. To assess model quality and increase confidence, XAI methods – in particular SHAP – as well as methods for analyzing data drift are integrated and available through both interfaces. Figure 1 provides an overview of the S4L platform, illustrating its key components.

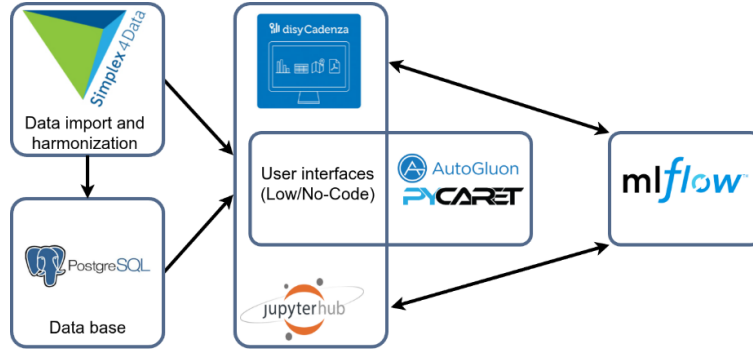


Fig. 1. Overview of the S4L platform showing its key components.

In addition to the functional components, the “SimplexTwIS” forms the foundation of the data architecture. It combines knowledge graphs and object hierarchies with a PostgreSQL database to ensure interoperability and machine-readable metadata. Data is provided through OGC-compliant services. The associated data processing pipeline comprises four essential steps, as illustrated in Figure 2.

1. **Data Import:** Import of domain-specific data and storage in the database component of the S4L platform.
2. **Data Harmonization:** Transformation of domain-specific data into a uniform, generic data model to ensure interoperability.
3. **Spatial Intersection and Enrichment:** Automated linking of station data with meteorological raster data based on the respective station coordinates – without manual intervention in the processing logic.
4. **Dataset Generation:** Consolidation of the harmonized data into a final dataset optimized for machine learning.

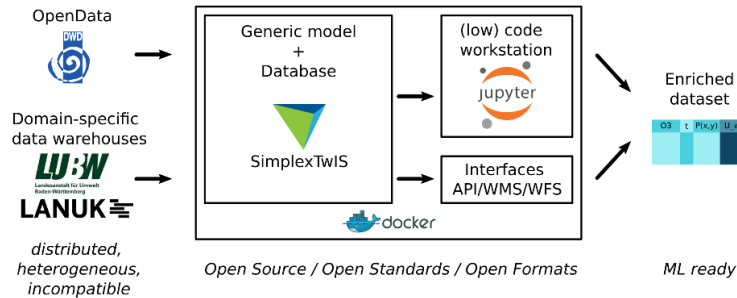


Fig. 2. Data Flow Architecture of the S4L Data Preparation Component on the S4L Platform

The methodological approach of S4L is based on an adapted version of the CRISP-DM process model [29]. A key difference from classical data mining projects is that the process does not necessarily begin with a specific business objective. Instead, use cases are identified in preliminary agency workshops [30]. A further essential upstream step is data preparation, in which environmental data is consolidated and harmonized in a data warehouse independently of the specific ML objective. This applies in particular to reusable data sources, such as meteorological data from the German Weather Service (DWD) or geospatial information. Since this phase therefore takes place prior to the actual “data understanding” stage, the subsequent “data understanding” phase involves merging and analyzing the specific environmental data relevant to the analysis with the already prepared data holdings. In addition, the modelling phase was conceptually subdivided into the selection of a learning model including its base configuration, and the actual learning phase of the system. Furthermore, the phase “Apply Model” was introduced, covering all aspects of MLOps – that is, deployment, operation, and monitoring – and ensuring that domain experts are also able to comprehend the models used and assess their quality. Figure 3 illustrates this adapted ML process.

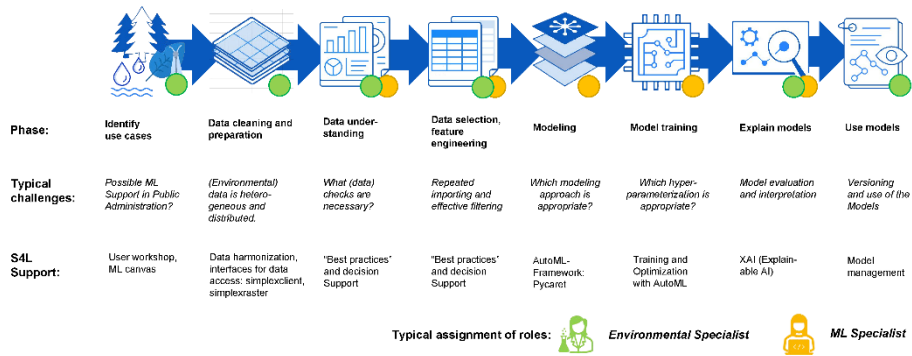


Fig. 3. Adapted ML pipeline of the S4L platform based on the CRISP-DM process model.

4 Methodology

The platform is evaluated using an environmental agency use case. The ML experiment uses a two-stage design to examine the platform's transferability between two German federal states (Baden-Württemberg, BW; Nordrhein-Westfalen, NRW), with ozone concentration prediction as the case study. The pipeline encompasses the entire process: data provision and preparation, feature engineering, modeling, and explainability. First, ML models are developed for three BW monitoring stations. Second, this identical pipeline is applied with unchanged parameters to an analogous problem in NRW. For each state, three station types: urban, industrial, and background—are included from the respective air quality monitoring networks.

The modeling differs only with respect to the station-specific air quality data. All other data sources – in particular meteorological data from the DWD – as well as the derivation of features are designed generically. The target variable is ozone concentration (O_3) with a forecast horizon of six hours. The features developed and the transformations applied are listed in Table 1. Modelling was performed using PyCaret 3.3.2 as the AutoML framework, with the objective of predicting O_3 ($t+6h$). The dataset was divided into a training period (2015–2021), a validation year (2022), and a test period (2023–2024). A Timeseries-Split strategy ($n=5$) was applied for model validation. Performance evaluation was based on the metrics R^2 , MAE, and RMSE, while model interpretation was carried out using SHAP values.

Table 1. Feature-Engineering

Feature Category	Variables	Transformation	Number of Features
Air Quality (Target)	O_3 in 6h (forecast)	None	-
Meteorology (Raw)	Air temp., radiation, wind, dew point, rel. humidity, air pressure, cloud cover, precipitation	None	8
Cyclic Encoding	Hour of day, day of year, wind direction, week of year	sin/cos transformation	8
Calendar Features	Day of week, season, weekend	Categorical / Binary	13
Lag Features O_3	O_3 t-1, t-2, t-3, t-6, t-12, t-24, t-48	Time lag	7
Lag Features Meteorology	Temp. (t-1,2,3,6,24), radiation (t-1,2,3,6), wind (t-1,2,3)	Time lag	12
Rolling Windows	O_3 & radiation (mean + max for 3h, 6h, 12h, 24h); temp. (mean for 3h, 6h, 12h, 24h)	Maxima and means	24
Radiation Accumulation	Cumul. radiation during the day	Daily sum	1
Interaction Features	Cloud cover $\times$ radiation, $\Delta temp_{24h}$	Product / difference	2

5 Case Study: Demonstrating Transferable Air Pollutant Forecasting

This chapter illustrates how domain experts without prior knowledge in machine learning (ML) can use the S4L platform to develop forecasting models for investigating environmentally relevant questions. The exemplary use case is “air pollutant

modelling for cause and scenario analysis”, which was identified in a user workshop with the associated partner – see Chapter 4 and [30].

5.1 Case study description

Air pollutants such as ozone (O_3), nitrogen dioxide (NO_2), and particulate matter ($PM_{2.5}$, PM_{10}) are subject to complex meteorological and emission-related influences that complicate their temporal forecasting. While NO_2 and particulate matter are primary pollutants released directly from sources, ground-level ozone is a secondary air pollutant. It forms in the atmosphere through photochemical reactions from precursor pollutants such as nitrogen oxides (from traffic and industry) as well as volatile organic compounds (VOCs) originating from biological processes, combustion, or industrial activities [31]. Data-driven ML models make it possible to capture these complex relationships purely on the basis of historical measurement and weather data and to develop station-specific forecasting models – without requiring explicit process knowledge. [32]

The ML objective derived from the above use case is: the development of a data-driven model for predicting ozone concentrations at individual monitoring stations on the basis of meteorological parameters and historical air quality data. This process is described in the following.

5.2 Data preparation and harmonization

Data preparation and harmonization are central steps in transforming heterogeneous environmental domain data into a uniform format suitable for training and inference. The S4L platform fully abstracts the underlying technical complexity of these processing steps. While complex procedures – such as the harmonization of heterogeneous data sources using “SimplexTwIS” and the spatial intersection with raster data – run automatically in the backend, environmental domain experts interact exclusively via an intuitive interface. This is provided by the “simplexclient” and “simplexraster” modules, which consistently encapsulate the complex processes. From the user’s perspective, interaction is thus limited to three essential inputs: a) the selection of monitoring stations, b) the definition of the observation period, and c) the specification of the desired meteorological parameters. The result of the data preparation is a consistent, ML-ready dataset that links domain-specific air quality measurement data with regionalized weather data from the DWD and is directly available for subsequent feature engineering.

For the present use case, air quality data from open data repositories for BW and NRW [33] were enriched with two raster datasets from the DWD (see Figure 4). The “Hostrada” dataset [34] offers a resolution of 1×1 km and an hourly temporal resolution from 1995 onwards, encompassing meteorological parameters such as cloud cover, wind speed and direction, air and dew point temperature, relative humidity, and air pressure. Additionally, the “Radolan” dataset [35] was incorporated for precipitation data, featuring an equivalent resolution of 1×1 km and providing hourly measurements from 2005 onwards.

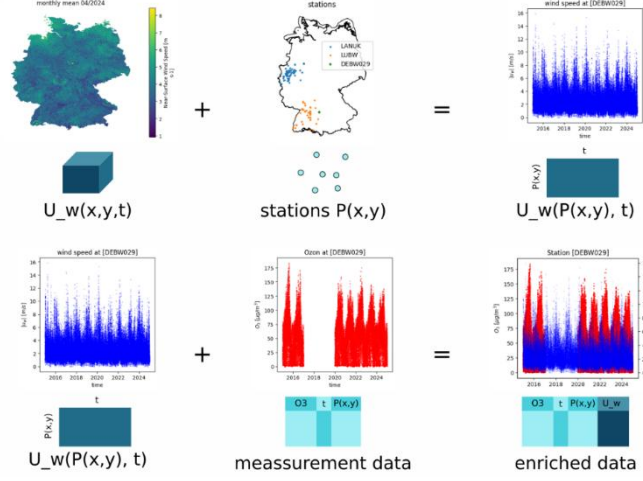


Fig. 4. Schematic representation of the transformations in the data preparation pipeline using the example of the air quality measurement data from station DEWB029 of the Baden-Württemberg monitoring network.

A key characteristic of this approach is its generic transferability. Since the spatial enrichment with DWD weather data is carried out in a fully coordinate-based and automated manner, the pipeline can be transferred to other federal states without manual adjustments; this is demonstrated using the example of the modelling for NRW in Chapter 5.7. Furthermore, the modular architecture, based on open-source software and open standards, enables the integration of additional external raster data sources – such as Sentinel satellite data – and thus a flexible expansion of the feature space in accordance with the requirements of the respective ML task.

5.3 Data Understanding and Feature Engineering

A thorough understanding of the input data is essential for the development of ML models. This encompasses in particular the analysis of data distributions, the identification and treatment of data gaps – for instance through appropriate interpolation methods – as well as correlation analyses to reduce highly correlated features. Since the quality of feature engineering substantially determines the predictive power of the models, a comprehensive literature review was conducted to identify and generically implement relevant features for the S4L platform. Based on this, the following feature classes – recommended in the specialist literature for air pollutants such as ozone, nitrogen oxides, and particulate matter – are available to environmental domain experts on the S4L platform for modeling ozone concentrations: a) meteorological parameters, b) temporal and spatial features, c) lag features, and d) rolling window features. In addition, specific features were implemented for ozone modelling that capture the photochemical formation from precursor pollutants. The features concretely used for modelling ozone concentrations are listed in Table 1 (Chapter 4).

The S4L platform thus implements generic yet curated methods to make frequently required features available to environmental domain experts without explicit ML expertise. The feature classes presented here relate to the current use case, but are transferable to other scenarios owing to the generalized approach. Furthermore, the use of regionalized weather data enables the implementation of complex, location-specific meteorological features. To support users in this regard, a preliminary literature review or expert consultation is necessary to identify and provide the appropriate feature methods. The feature engineering process concludes with the creation of a data table containing domain-specific subject data, regionalized weather data, and additionally computed complex features..

5.4 Implementation of the AutoML Workflow

Model development was done using the open-source AutoML framework PyCaret within a Docker environment. The selection of PyCaret was largely determined by the requirement expressed by domain experts in user workshops to focus on interpretable models rather than "black-box approaches" [28]. PyCaret automates essential steps of the modelling process – including data preparation, model comparison, hyperparameter optimisation, and model evaluation [36] – thereby enabling users without in-depth ML knowledge to independently carry out the entire modelling workflow. Manual interaction is essentially limited to the initial configuration of the framework.

As part of the automated model comparison, the regression algorithms available in PyCaret are evaluated, including linear models (e.g. Ridge, Lasso, ElasticNet), decision tree-based ensemble methods (e.g. Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost), as well as further non-parametric methods [36]. The model with the best performance in cross-validation is subsequently subjected to hyperparameter optimisation using Grid Search. Upon completion of training, the resulting model is automatically saved in the model management system MLflow for reuse or future model comparisons [37].

To support environmental domain experts in modelling ozone concentrations, a preconfigured "best practice" setting for PyCaret is provided, which can be modified for specific ML experiments. Although PyCaret largely automates the modelling process, a well-founded configuration still requires ML expertise. To ensure transferability to further use cases, a preliminary literature review or expert consultation is therefore necessary in order to define appropriate "best practice" configurations. The modelling configurations used for the present use case are documented in Chapter 4.

5.5 Model Interpretability

To make the decision-making processes of the developed ML models transparent and comprehensible, and to address domain-specific questions, Explainable AI (XAI) in the form of SHAP values has been implemented in the S4L platform. This enables a physically interpretable diagnosis of model deviations and thus provides a well-founded basis for the domain-specific validation of model results. Since SHAP values

are based on the variation of input parameters, they are model-agnostic and require no knowledge of the internal model structure [7].

Figure 5 illustrates, by way of example, the feature importance of a model for predicting ozone concentration. The five most influential features (Top-5) are displayed in the form of SHAP violin plots. A supplementary explainer dashboard additionally offers further visualization options as well as the possibility for interactive exploration of the results. This enables environmental domain experts to analyze model decisions directly in the context of their domain-specific questions.

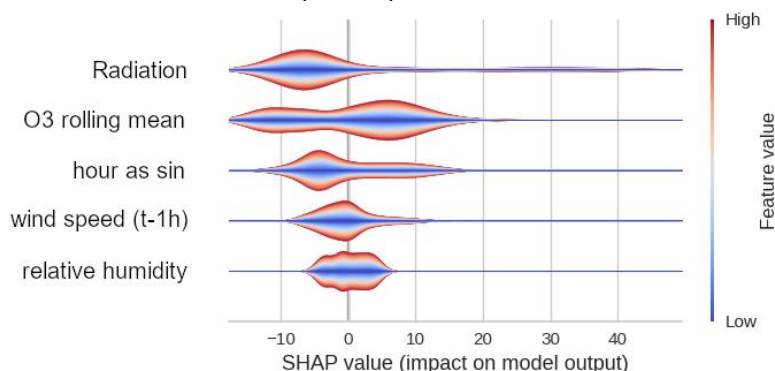


Fig. 5. SHAP violin plots of the Top-5 features for predicting ozone concentration. The color coding represents the magnitude of the respective feature values.

5.6 Modelling Results

Predictive ozone models for BW were developed and evaluated across three station types: urban, industrial, and background sites. As shown in Figure 6, the models accurately reproduce measured ozone maxima and exhibit consistent performance across all sites. Model evaluation and ranking were automated using PyCaret; the top-performing models are detailed below. On the test dataset, the Light Gradient Boosting Machine achieved an R^2 value of 0.79 for the urban station Karlsruhe-Nordwest (MAE: 10.40 $\mu\text{g}/\text{m}^3$; RMSE: 13.71 $\mu\text{g}/\text{m}^3$). For the industry-adjacent near station, the same model yielded an R^2 value of 0.81 (MAE: 10.38 $\mu\text{g}/\text{m}^3$; RMSE: 13.57 $\mu\text{g}/\text{m}^3$). At the background station Schwäbische Alb, the Extra Trees Regressor achieved an R^2 value of 0.77 (MAE: 8.87 $\mu\text{g}/\text{m}^3$; RMSE: 11.45 $\mu\text{g}/\text{m}^3$). The R^2 values (0.77 to 0.81) demonstrate solid model quality and generalizability of the chosen AutoML approach. The MAE and RMSE values indicate that not all emission events are fully captured; however, they fall within a range that permits further analyses. To ensure reproducibility, all modelling runs are versioned using MLflow; in addition, XAI methods (see Section 5.5) enable an examination of model decisions. Overall, the developed models exhibit sufficient quality for use in scenario comparisons.

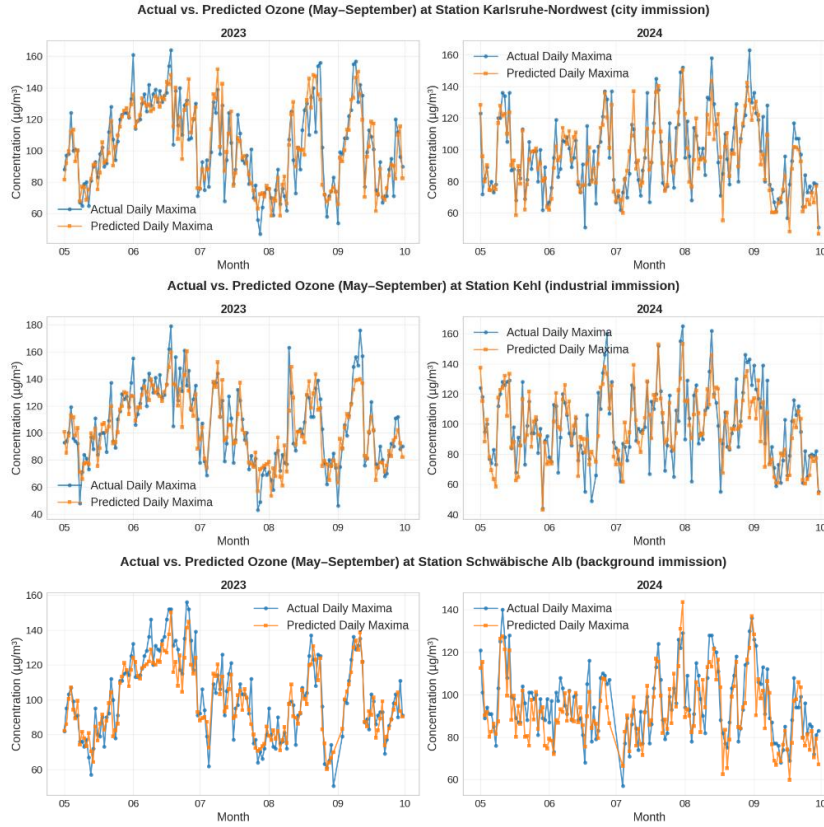


Fig. 6. Comparison of measured daily ozone maxima and modelled 6-hour forecasts for three station types: urban, industrial, and background sites in Baden-Württemberg (2023–2024).

5.7 Transferability to Other Federal States

To demonstrate the transferability of the modelling pipeline, analogous modelling was carried out for three monitoring stations in NRW. These were selected according to the same criteria as in BW (urban, industrial, and background sites). Figure 7 shows the ozone maxima during the test period for each station. While the models for the urban and industry-adjacent stations achieve comparable quality, the background station deviates. The results of the best-performing models are presented below. On the test dataset, the Light Gradient Boosting Machine achieved an R^2 value of 0.72 for the urban station Münster-Geist (MAE: $11.83 \mu\text{g}/\text{m}^3$; RMSE: $15.25 \mu\text{g}/\text{m}^3$). For the industry-adjacent station Niederzier, the same model yielded an R^2 value of 0.70 (MAE: $10.75 \mu\text{g}/\text{m}^3$; RMSE: $13.97 \mu\text{g}/\text{m}^3$). At the background station Simmerath (Eifel), Linear Regression achieved an R^2 value of 0.59 (MAE: $10.58 \mu\text{g}/\text{m}^3$; RMSE: $13.94 \mu\text{g}/\text{m}^3$). The results for the three NRW stations show that comparable, station-specific predictive models could be generated for the urban and industry-adjacent stations. Regarding the background station, it should be noted that 6.14 % of ozone

measurements were missing (1.59 % remaining after interpolation). As shown in Figure 7, this data loss has a negative impact on predictive quality.

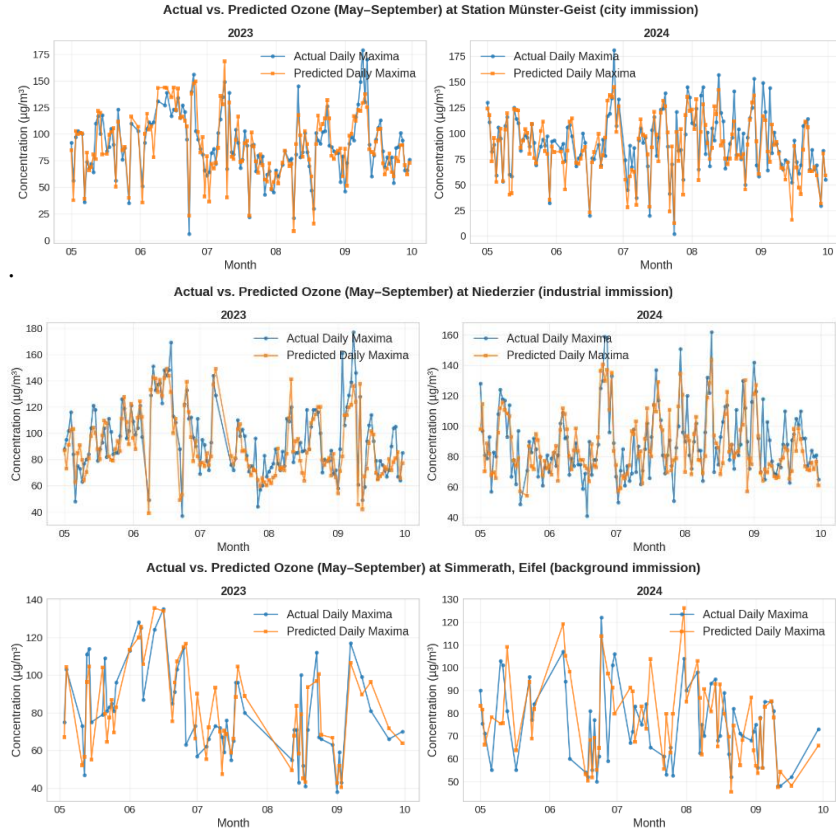


Fig. 7. Comparison of measured daily ozone maxima and modelled 6-hour forecasts for three station types: urban, industrial, and background sites in NRW (2023–2024).

6 Discussion

The S4L platform presented here was designed with the goal of lowering barriers to the use of machine learning methods in public environmental administration. It aims to make these methods accessible to domain experts without in-depth machine learning expertise. The platform addresses three complementary core objectives:

- Establishing a uniform, harmonized data foundation,
- Practical and accessible application for domain experts,
- Ensuring transferability.

The following discussion addresses the results presented in Chapter 5 in the context of these objectives, the technical implementation steps, and practical suitability.

The systematic data preparation of domain and DWD raster data was initially implemented for a specific use case. However, the underlying methodology can be applied to other environmental data, enabling efficient enrichment with meteorological parameters and platform-wide reusability. However, it must be noted that the initial data preparation and schema definition require IT expertise and must therefore be carried out by a specialized data professional (“Creator role”, see also [28]).

Regarding data understanding and feature engineering, the implemented methods are designed generically. The tools for exploratory data analysis can be readily transferred to other use cases; initial successful integrations have already been realized in the established analysis platform Disy Cadenza. While standard feature engineering methods – such as the automatic generation of lag features, calendar features, or rolling windows – are generically implemented and thus directly transferable, automated approaches reach their limits with highly domain-specific features and continue to require manual implementation.

The PyCaret framework integrated into the platform consolidates the most important algorithms for regression, time series analysis, classification, and clustering within a unified interface. This makes the technical pipeline applicable to a broad spectrum of administrative use cases. A further central component is the seamless traceability of ML experiments via MLflow.

The explainable AI (XAI) methods integrated into S4L – in particular those based on SHAP values – enable environmental professionals without in-depth ML expertise to comprehend the basis of a forecast and assess its plausibility. Conceptually, it is also planned to visualize model inference directly via Cadenza in the future.

The model results achieved for the specific use case – air pollutant forecasting for scenario analysis – demonstrate sufficient accuracy for the intended purpose. While more precise models for air pollutant forecasting exist in the scientific literature, it is important to note that the models presented here are produced within a standardized, automated process. This enables environmental professionals to train, adapt, and apply the models independently and without programming effort. The integrated XAI methods (SHAP) additionally provide a visually intuitive representation of feature contributions, allowing environmental professionals without deep ML expertise to conduct well-founded model evaluations. As SHAP methods are model-agnostic, they can be applied uniformly across a wide variety of model types.

It was demonstrated that the modelling can be successfully transferred to other federal states through the standardized, automated process. Environmental professionals are thus able to apply the described workflow independently to the data of their respective regional air quality monitoring networks, without requiring any modifications to the underlying platform architecture or model configuration. A prerequisite for transferability is that the respective measurement data have been integrated into the data model on a one-time basis.

In its current implementation, the S4L platform is not mature enough for broad deployment in administrative practice. While the methodological and technical feasibility has been demonstrated using the example of the use case presented, the platform remains limited to specific, curated scenarios. A significant barrier to operation is the heterogeneous level of ML expertise within public administration. This challenge

must be addressed through platform-level support, which is already implemented in the S4L platform through guided workflows and a strict division of roles.

7 Summary and outlook

Environmental authorities face the challenge of utilizing complex data for well-informed decision-making without having access to requisite machine learning expertise. Existing AutoML solutions insufficiently address the specific requirements of public administration – such as end-to-end workflow support and generalizability.

The Simplex4Learning project addresses this gap with an ML platform that enables professionals in environmental authorities to conduct domain-specific analyses without in-depth ML knowledge. The platform combines methods of Automated Machine Learning (AutoML) and Explainable AI (XAI) with efficient approaches to (geo-)data integration and management. This enables a user-friendly, effective, and efficient application of machine learning methods in environmental administration. Conceptually, the S4L platform is not restricted to a single domain and can in principle also be deployed outside the environmental sector.

This paper outlines the S4L platform and demonstrates its functionality using air pollutant forecasting. A key characteristic of this transfer study is the minimal adaptation effort required: the entire modelling pipeline – including feature engineering, AutoML configuration, and SHAP-based model interpretation – remained unchanged. Only the input data were exchanged, replacing the measurement time series from the Baden-Württemberg air quality monitoring network with the corresponding data from NRW. The data pipeline automatically enriched the dataset with regionalized weather data from the DWD based on station coordinates. This automation eliminated the need for manual intervention in the processing logic. The discussion addressed the current state of development of the platform as well as its practical suitability for administrative use.

It is planned to further develop parts of the platform into a product and integrate them into Cadenza. A more extensive domain focus will develop organically through practical use. As usage increases, the ML repository grows and provides an expanding number of reusable models for diverse environmental questions. In future development stages, this aspect could be further deepened through the implementation of environment-specific controlled vocabularies for model description.

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Competing Interests

The authors declare that they have no competing interests.

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